We will now see that the information about f' obtained from the Mean Value Theorem is strong enough to allow us to deduce information about f that so far has been out of reach. These results will play an important role in the rest of the course. We begin by proving the converse of the theorem that if f is constant, then $f'(x) = 0$ for all x.

Corollary 1. If f is defined on an interval and $f'(x) = 0$ for all x in the interval, then f is constant on the interval.

1. Prove Corollary 1 by showing that $f(a) = f(b)$ for all points a, b in the interval. *Hint: Choose any two points a and b on the interval with $a \neq b$ and apply the Mean Value Theorem to f on $[a,b]$.*

2. Consider the function

$$f(x) = \begin{cases} 1, & \text{if } x > 0, \\ -1, & \text{if } x < 0. \end{cases}$$

We see that $f'(x) = 0$ everywhere f is defined, but f is not constant. Is this consistent with the proposition above? Explain.

The next result shows that a function defined on an interval is uniquely determined by its derivative, up to the addition of a constant function.

Corollary 2. If f and g are defined on the same interval, and $f'(x) = g'(x)$ for all x in the interval, then $f - g =$ constant.

3. Prove Corollary 2. *(Hint: Apply Corollary 1 to the function $h = f - g$.)*

Definition. An *antiderivative* of a function f defined on an interval I is a function F such that $F' = f$ on I. For example, $\frac{x^3}{3}$ is an antiderivative of x^2 since on any interval we have

$$\left(\frac{x^3}{3}\right)' = \frac{1}{3}(x^3)' = \frac{1}{3}\cdot 3x^2 = x^2.$$

By Corollary 2, any other antiderivative of x^2 must be of the form $\frac{x^3}{3} + C$, where C is a constant.

Antiderivatives will play an important role in the calculation of integrals in BC 2.

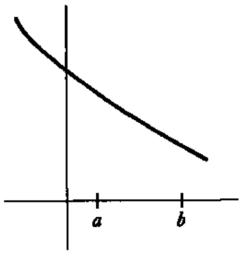
4. Find the general form for an antiderivative for the function $f(x) = \cos(x)$.

For the next corollary, recall the following definitions:

Definition. (Increasing, decreasing functions) A function f is *increasing* on an interval if $f(a) \leq f(b)$ whenever a and b are two numbers in the interval with $a < b$. The function f is *decreasing* on an interval if $f(a) \geq f(b)$ for all a and b in the interval with $a < b$.



(a) An increasing function.



(b) A decreasing function.

It is easy to see that if f is differentiable and increasing on an interval, then $f'(x) \geq 0$ for all x in the interval: for if $h \neq 0$, then since f is increasing we have for each x in the interval $\frac{f(x+h)-f(x)}{h} \geq 0$, and it follows that $\lim_{h\to 0} \frac{f(x+h)-f(x)}{h} \geq \lim_{h\to 0} 0$ and therefore $f'(x) \geq 0$. [1] Similarly, if f is differentiable and decreasing on an interval, then $f'(x) \leq 0$ for all x in the interval.

What about the converse of these statements? That is, if $f'(x) > 0$ on an interval, is f necessarily increasing at every point of the interval? And if $f'(x) < 0$ on an interval, is f necessarily decreasing at every point of the interval? We now use the Mean Value Theorem to show that this is indeed the case. **Corollary 3.** If $f'(x) > 0$ for all x in an interval, then f is increasing on the interval; if $f'(x) < 0$ for all x in the interval, then f is decreasing on the interval.

5. Prove Corollary 3. *(Hint: For each sign of $f'(x)$, choose two points a and b in the interval with $a < b$ and apply the Mean Value Theorem to f on $[a,b]$ to show that $f(a) < f(b)$.)*

[1] We have used the fact that if $f(x) \leq g(x)$ for all points on an interval containing a except perhaps at a itself, and if $\lim_{x\to a} f(x)$ and $\lim_{x\to a} g(x)$ both exist, then $\lim_{x\to a} f(x) \leq \lim_{x\to a} g(x)$. This is easily proved by contradiction (suppose $\lim_{x\to a} f(x) > \lim_{x\to a} g(x)$ and take ε to be half their difference).

For the first two corollaries, the converse is also true. Is the converse of Corollary 3 true?

6. If f is increasing on an interval, must we have $f'(x) > 0$ for all x in the interval? Give a proof or counterexample. *(Hint: Consider $f(x) = x^3$.)*

Corollary 3 provides enough information to get a good idea of the graph of a function with a minimal amount of point plotting. We will have a lot more to say about how f' affects the graph of a function in the next lesson.

7. Consider the function $f(x) = x^3 - 3x$. Use Corollary 3 to find the intervals on which f is increasing and the intervals on which f is decreasing.

♢ **Check your work with your instructor.**